

Shreya Lakshmisai Maddhali

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University of California, Davis: Major in Computer Science, Minor in Statistics

Relevant Coursework: Object-Oriented Programming, Data Structures & Algorithms, Computer Architecture, Computer Vision, Agent-Based Modeling, Probability and Statistical Modeling, Technical Writing

Technical Skills: Git, Programming in Java, Python (Numpy, Pandas, Matplotlib, Seaborn, PyTorch, TensorFlow), C, HTML, R, Machine Learning, Artificial Intelligence, ServiceNow (ISTM), SQL, Kotlin, Project Management, ticketing systems, developing and maintaining dashboards

Soft skills: Customer service, excellent written/verbal communication skills, hospitality, call center support

Certifications/Trainings: CompTIA A+ Core 1, UC Cyber Security Fundamentals, CITI Program: Biomedical Researchers and staff & GCP for Clinical Trials with Investigational Drugs and Biologics (ICH Focus)

Technical Experience + Projects

IT Analyst, UC Davis College of Letters and Science

February 2024 - Present

- Provided technical support for 1,000+ faculty and staff across 30+ departments, resolving 6,000+ service requests via phone, walk-in, and ticketing system while maintaining over 4,000 Windows and macOS devices in a high-volume higher-education IT environment.
- Troubleshoot wireless and network connectivity issues (eduroam, campus Wi-Fi, Ethernet, and access points) by diagnosing configuration errors, verifying physical connections, and coordinating with network teams to restore service with minimal downtime.
- Worked in Unix/Linux (Ubuntu) environments using command-line tools, including grep, pipes, bash utilities, apt package management, and sudo, to debug system, software, and configuration problems.
- Monitored network and server status using Site247, responding to alerts, performing initial diagnostics, and escalating incidents to system and network administrators to ensure continuous operation of campus IT services.
- Configured and maintained static IP assignments, printers, network access modules (NAMs), and remote-access restrictions, supporting secure connectivity for offices, labs, and departmental systems.
- Installed, imaged, and managed devices using Jamf, BigFix, and OpenVPN, including enrollment, permissions configuration, software deployment, and patch updates to maintain cybersecurity compliance.
- Assisted users with phishing prevention, account security, and access issues, supporting services such as Exchange, Gmail, Box, Dropbox, and university authentication systems.
- Created internal and user-facing technical documentation for troubleshooting, network setup, and device configuration, improving response time for recurring issues and supporting both technical and non-technical users.

Neuropathology Core Research Assistant - UC Davis Health, Dugger Laboratory

July 2024 - Present

- Collaborating cross-functionally with neuropathologists, engineers, and data scientists to align laboratory workflows with digital AI segmentation pipelines.
- Digitally annotated 20+ postmortem human brain Whole Slide Images (WSIs) in QuPath to identify gray matter, white matter, and leptomeningeal regions for machine learning training sets.
- Implemented quality control and preprocessing protocols in Python, ensuring 99% dataset accuracy and improving segmentation model reproducibility for dementia-related biomarker detection.
- Supported over 10+ grossing procedures, assisting with anatomical orientation, biospecimen labeling, and metadata archiving for Alzheimer's Disease Research Center (ADRC) cases.
- Contributed to curriculum learning-based fine-tuning frameworks for parameter-efficient segmentation, improving leptomeningeal classification accuracy by over 30% IoU across WSIs.
- Integrated segmentation models with Amyloid- β plaque detection pipelines, advancing region-specific quantification for cerebral amyloid angiopathy (CAA) analysis.
- Managed inventory, documentation, and database organization to support NIH-funded grant deliverables and cross-lab data harmonization.

Project: Curriculum Learning Approach for Enhanced Brain Segmentation in Whole Slide Images (IEEE Access, 2025)

- Contributed annotated datasets and experimental support for a curriculum learning (CL) and parameter-efficient fine-tuning (PEFT) framework designed to accelerate segmentation model retraining.
- Designed the annotation pipeline using QuPath and Python preprocessing scripts to generate leptomeningeal masks used for training and evaluation.
- Contributed to a project that achieved a +13.5% IoU and +15% Dice improvement on the SegFormer backbone, demonstrating robust model adaptation across stains.
- Co-authored the publication under the mentorship of Dr. Brittany N. Dugger and Dr. Chen-Nee Chuah, contributing to dataset preparation and experimental support.

Project: Image Generation using U-NET Architecture (2024)

- Designed an innovative custom loss function that improved mapping accuracy between RGB and multispectral image datasets; findings led directly to strengthened performance on two key agricultural data sets analyzed during project execution
- Delivered a comprehensive codebase and a summary report, positioning the model as a cost-effective alternative to traditional multispectral cameras

Leadership and Management

Treasurer, Women in Computer Science, Davis Chapter

May 2024 - Present

- Secured \$20,000 in sponsorships from local businesses through strategic outreach efforts; funds directly supported new initiatives aimed at increasing member engagement and expanding educational programs within the community

Career and Development Team Co-lead, Davis Data Science Club

May 2024 - Present

- Facilitated a series of 5 data science workshops and 3 networking events that fostered connections among 75+ members, resulting in improved collaboration on projects and enhanced knowledge-sharing within the community

Chief Financial Officer & Sponsorship Lead, HackDavis

September 2023 - January 2025

- Directed organization of all financial documentation in Excel spreadsheets for HackDavis; streamlined reporting processes by saving 8 hours weekly on management tasks, raised over \$25,000 in sponsorships from companies and nonprofits

Publications

- Histo-Hallmark: Hallucination Benchmark Suite for Large Vision-Language Models in Histopathology Tasks. AAAI 2026 AI Alignment Track Submission — Co-author.
- Curriculum Learning Approach for Enhanced Brain Segmentation in Whole Slide Images with Fast Labeling. IEEE Access, 2025 — Co-author.